Cleaning and Merging Datasets

Methods

Human genetic sampling was conducted under informed consent procedure approved by the Pennsylvania State University and the Madagascar Ministry of Health. Participants were recruited from five localities in Madagascar, and each participant was asked their ethnic group and that of their grandparents. Saliva containing DNA was collected from each participant using Oragene collection tubes. DNA was extracted from saliva samples, and DNA concentrations were standardised in preparation for genotyping using Nanodrop.

Genetic and phenotypic data were collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands (n = 36), Betsileo people from the southern Highlands (n = 38), Sakalava and Tsimihety people from the northwestern coast (n = 40), Betsimisaraka people from the eastern coast (n = 38), and people of various ethnic groups from the extreme north in the town of Diego (n = 40). ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global comparative dataset. The Malagasy dataset was created by merging our data with Malagasy data from Pierron et al. (2014) (Mikea; n = 21, Temoro; n = 24, Vezo; n = 24). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the Auton et al. (2015) dataset (n = 882), Bantu African individuals from Fortes-Lima et al. (2024) (n = 1763), individuals from Borneo from Brucato et al. (2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from the full dataset of 240 individuals) and Timor (n = 7) Cox et al. (2016). The genome build of each dataset was verified using the function `snp_modifyBuild` from `bigsnpr` Privé et al. (2018) prior to merging to ensure each dataset used the same build (in this case, GRC37).

Data were merged in R using tidypopgen function `rbind` Carter and Manica (2025), which merges SNP datasets, resolves strand inconsistencies, and removes ambiguous SNPs. After merging, datasets were cleaned using tidypopgen Carter and Manica (2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.05 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. SNPs with minor allele frequency below a threshold of 0.025 were removed. Hardy-Weinberg equilibrium p-values were calculated for each SNP within each population using the tidypopgen function qc_report_loci, which provides a Bonferroni correction given the number of populations analysed. SNPs with a p-value below 0.01 in at least one population after correction were removed. Close relatives were removed using the KING robust kinship coefficient Manichaikul et al. (2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 596146 SNPs, and the global dataset consisted of 2744 individuals and 425931 SNPs.

SNPs in linkage were then removed by LD clumping Privé et al. (2018), using tidypopgen. Those with an r2 value greater than 0.2 in a window of 100 kilobases were removed. After LD clumping, the Malagasy dataset consisted of 227410 SNPs and the global dataset consisted of 198196 SNPs.

Principal components analysis (PCA) was used firstly to visualise the Malagasy dataset, and secondly the Malagasy individuals compared to global comparative data. Non-malagasy populations were randomly subsampled to a maximum of 30 individuals to avoid bias through uneven sample sizes. This left a total of 1955 individuals.

Unsupervised clustering using the model-based method fastmixture Santander, Refoyo Martinez, and Meisner (2024) was run on the full, cleaned dataset of 2744 individuals and 425931 SNPs using the package tidygenclust Tysall et al. (2025). Fastmixture has been shown to be computationally efficient for large datasets while providing comparable results to other model-based clustering algorithms such as STRUCTURE Pritchard, Stephens, and Donnelly (2000) and ADMIXTURE Alexander, Novembre, and Lange (2009), with reported speeds up to 30 times faster than ADMIXTURE Fumagalli (2024).

The algorithm was run for 9 values of K (2 to 10) and 3 runs per K with 5-fold cross validation. Subsequently, we analysed the fastmixture output using the Clumppling algorithm of Liu, Kopelman, and Rosenberg (2024), also implemented in tidygenclust Tysall et al. (2025). Clumppling aligns replicated runs of unsupervised clustering across multiple values of K to identify modes across replicates. A single mode was found for each value of K across 3 replicates.

f_2 statistics were calculated in tidypopgen Carter and Manica (2025), f_3 and f_4 ratio statistics were then calculated in Admixtools2 from these f_2 blocks Maier and Patterson (2024). Admixture f_3 statistics were used in the configuration f_3(Malagasy; Asian, African) to test whether the Malagasy population could be modelled as an admixture between the given African and Asian references.

As the Merina exhibited high inter-individual variation in fastmixture ancestry profiles, indicative of some form of population structure, we ran the supervised mode of ADMIXTURE Alexander, Novembre, and Lange (2009) at K = 2 for all Malagasy populations. Individuals from Borneo and individuals from Mozambique were used as reference populations in this supervised mode, as these reference populations were shown to give the minimum admixture f3 statistics values. Every Malagasy population showed inter-individual variation in ancestry proportions. However, supervised ADMIXTURE profiles revealed a bimodal distribution among the Merina Alexander, Novembre, and Lange (2009), which has been observed previously in this population Hodgson (2016). Merina with a high proportion of African ancestry and the Merina with a high proportion of Asian ancestry were then separated into two groups before running MALDER and pairwise FST analyses. Each individual was assigned to group Merina A or group Merina B based on their ancestry profile at k = 2 inferred using supervised ADMIXTURE Alexander, Novembre, and Lange (2009), with Merina A estimated to have >50% of ancestry derived from a population similar to Borneo individuals, and Merina B estimated to have >50% of ancestry derived from a population similar to Mozambique individuals. See supplementary Figure 2 for results of supervised ADMIXTURE analysis. Splitting this sample reduced power, as only 12 individuals remained in the higher African ancestry group. However, doing so avoided violating model assumptions (e.g the model of linkage employed by ALDER assumes that there is no structure within populations, that individuals are mating at random) and enabled us to consider whether there may be differing demographic histories among the Merina.

f_4 ratio statistics were used to examine the proportion of ancestry derived from Asian versus African sources for each population. To obtain these ratios, the global dataset was merged with 1000 Genomes GBR and a chimpanzee reference (PanTro3.0, GenBank accession GCA_000001515.5) to act as outgroups. f_4 ratios were calculated in the configurations f_4(GBR,Chimpanzee;Malagasy,Borneo) / f_4(GBR,Chimpanzee;Esan,Borneo) and f_4(GBR,Chimpanzee;Malagasy,Esan) / f_4(GBR,Chimpanzee;Borneo,Esan).

To obtain a chimpanzee reference, first SNPs in our dataset were lifted to PanTro3.0 (GenBank accession GCA_000001515.5) from hg19 using liftOver overchain file https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro5.over.chain.gz from UCSC Genome Browser Kent et al. (2002), through bigsnpr function snp_modifyBuild() Privé et al. (2018). The chimpanzee reference for each SNP was then obtained using a custom R function to query the Ensembl API (GenBank accession GCA_000001515.5, Ensembl release 115, accessed via Ensembl API https://rest.ensembl.org/documentation/info/sequence_region on 17-12-2025). This procedure retained 77993 SNPs for f_4 ratio calculation. Standard errors were calculated using block-jackknife resampling (admixtools f_4 ratio default). See Supplementary Table 3 for full f_4 ratio output.

MALDER Loh et al. (2013); Pickrell et al. (2014) was used to date the timing of admixture across different Malagasy groups. Genetic distances for each SNP were set using HapMap Phase II genetic map Frazer et al. (2007), mapped to build GRCh37. tidypopgen was used to create a subset of the full dataset containing only SNPs with known genetic map positions in the correct order. This dataset was not pruned for linkage

disequilibrium, and consisted of 236715 SNPs. MALDER was run using the default -binsize of 5 * 10-4 Morgans (describing the size of the bins that SNPs are split up into. Instead of directly calculating distance between each SNP pair, MALDER uses the distance between the bins of each SNP pair.), and the –mindis parameter (minimum genetic distance to begin LD decay curve fitting) set to 0.005 cM. Standard error of estimated dates were calculated through leave-one-chromosome-out jackknifing, with the flag -jackknife.

Pairwise F_{st} distances between Malagasy populations were calculated in tidypopgen, using the method of Weir and Cockerham (1984).

Supplementary Data

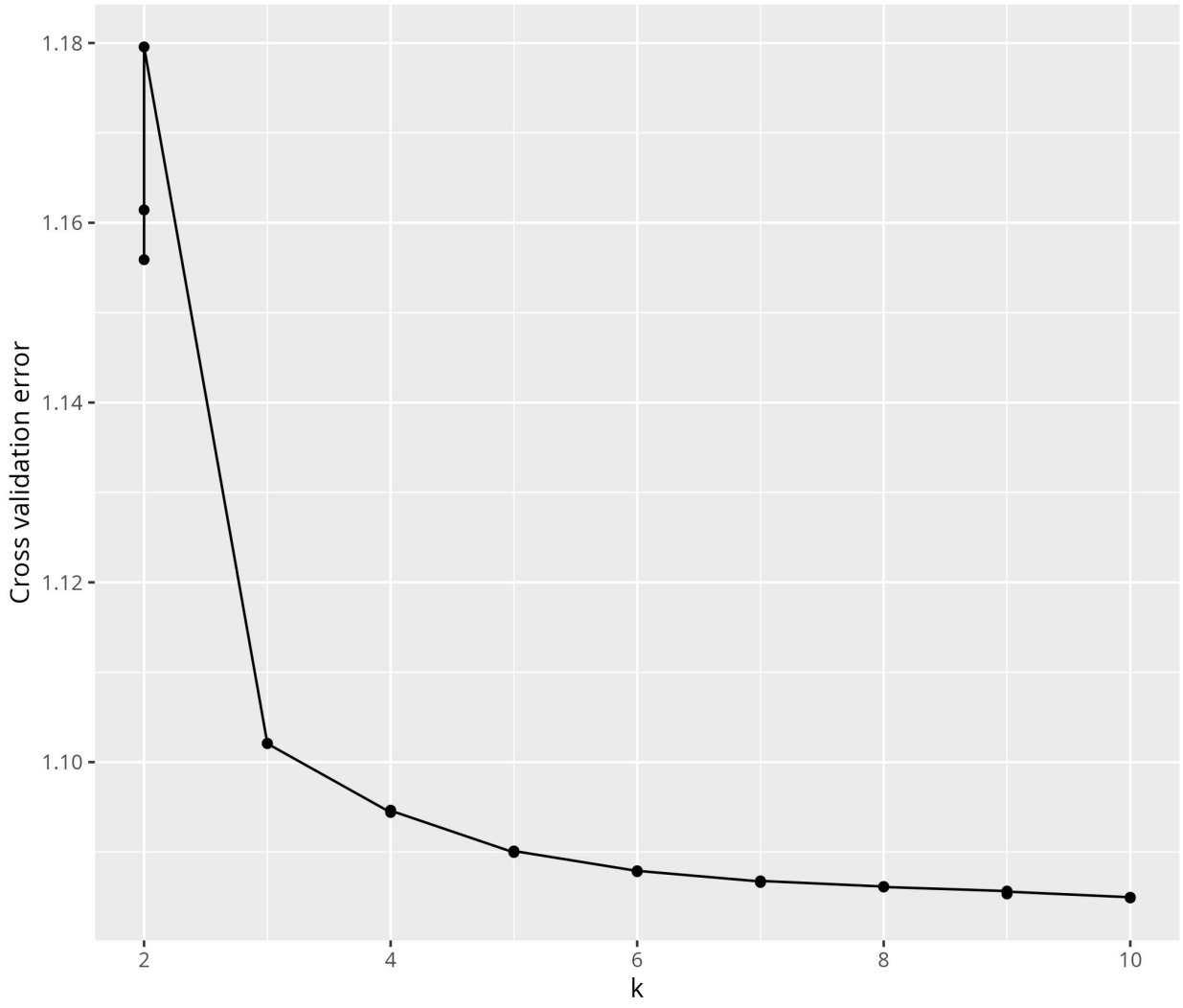


Figure 1: Supplementary Figure 1: Cross-validation error for fastmixture.

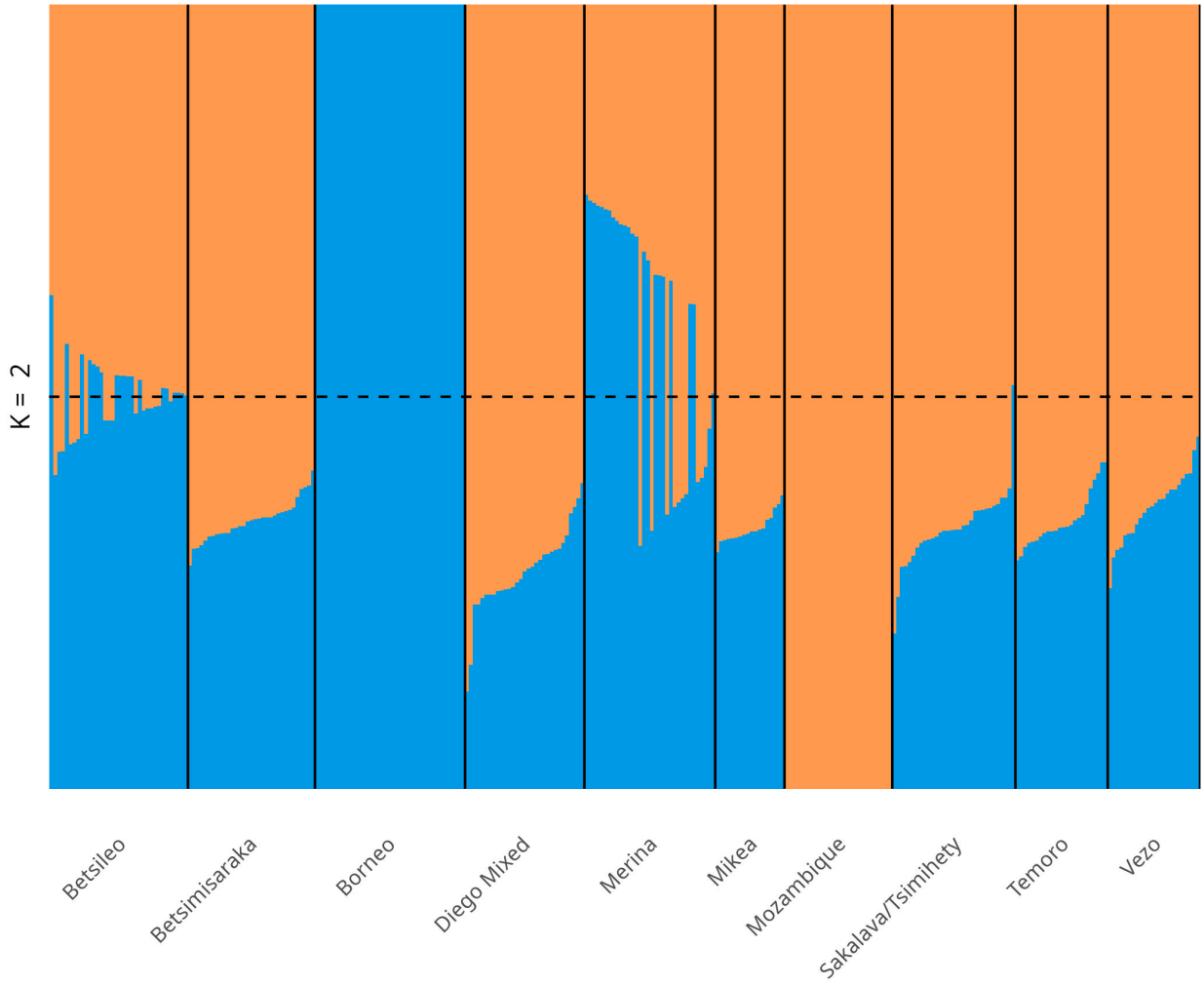


Figure 2: Supplementary Figure 2: Supervised admixture at K = 2 for for Malagasy compared to reference populations. Individuals from Borneo from Brucato et al. (2016), and individuals from Mozambique populations from Fortes-Lima et al., (2024). Mozambique individuals are labelled together, but this group contains individuals given the following population labels: “Mozambique_Nhungwe”, “Mozambique_Tsonga”, “Mozambique_Ronga”, “Mozambique_TongaMoz”, “Mozambique_Chuwabu”, “Mozambique_Tswa”, “Mozambique_Chopi”, “Mozambique_Sena”, “Mozambique_Makhuwa”.

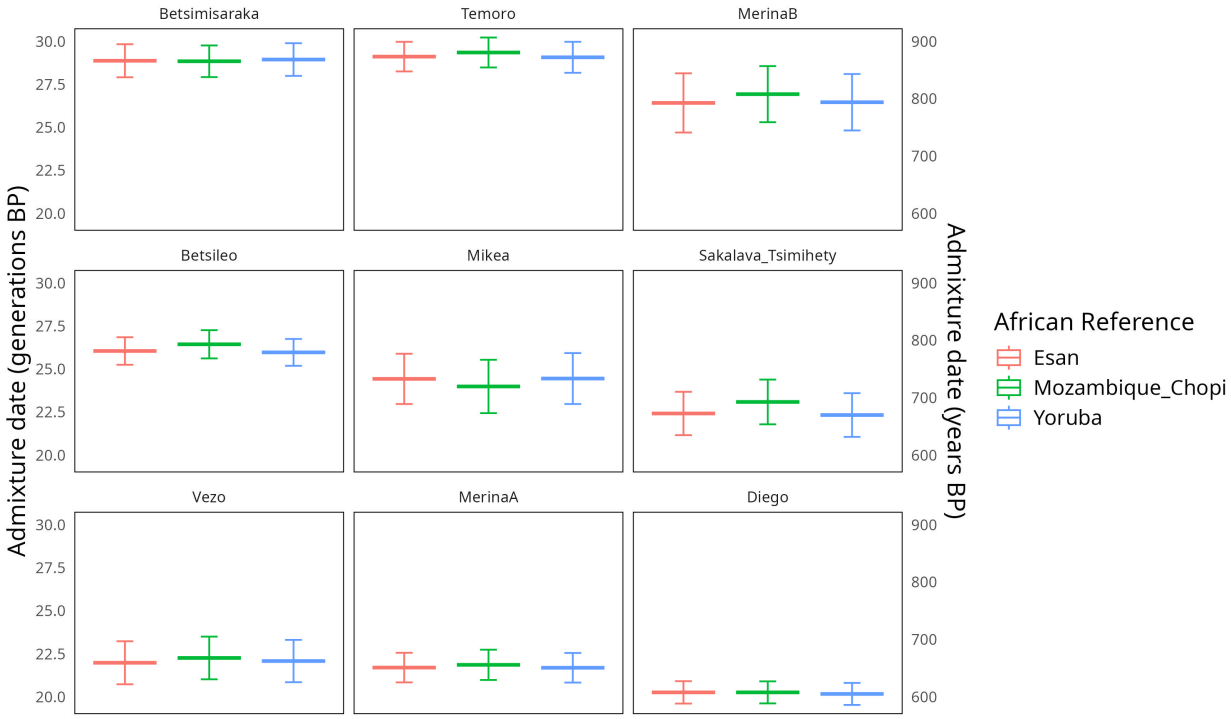


Figure 3: Supplementary Figure 3: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Borneo as an Asian source population and varied African source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

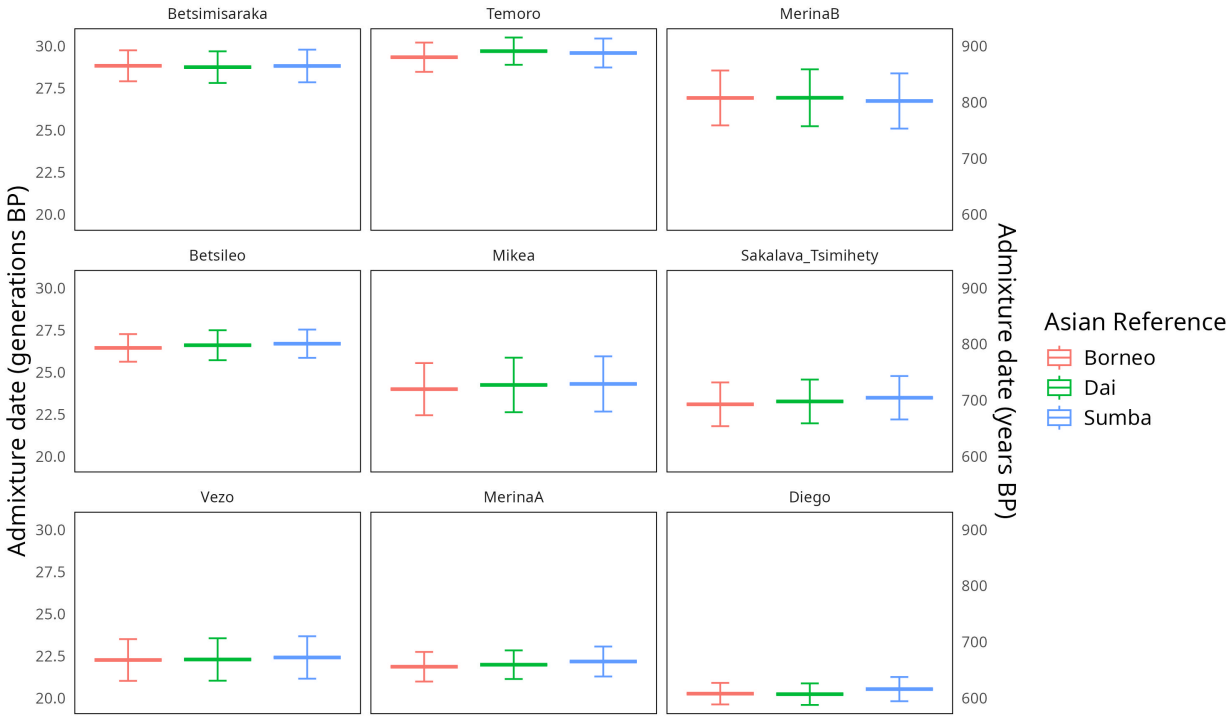


Figure 4: Supplementary Figure 4: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Mozambique as an African source population and varied Asian source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

Table 1: Supplementary Table 1: Summary of analyses and software used in this study.

Analysis	Package/function	Dataset	Individuals Removed	n indivs	Loci filtering	n SNPs
Malagasy PCA	tidypopgen: gt_pca_randomSVD	Malagasy dataset	Missingness > 0.1; KING kinship > 0.0884	233	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	227410
Global PCA	tidypopgen: gt_pca_randomSVD	Global dataset	Missingness > 0.1; KING kinship > 0.0884	2744	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	198196
Unsupervised clustering	tidypopgenclust: gt_fastmixture	Global dataset	Missingness > 0.1; KING kinship > 0.0884	2744	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	198196
Supervised clustering	ADMIXTURE: ADMIXTURE	Subset of Global dataset	Missingness > 0.1; KING kinship > 0.0884	299	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	198196
F3 statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qp3pop	Global dataset	Missingness > 0.1; KING kinship > 0.0884	2744	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	198196
F4 ratio statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qpf4ratio	Global dataset + Pan and European outgroups	Missingness > 0.1; KING kinship > 0.0884	2792	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	77993
MALDER	MALDER (version 1.0) (Pickrell et al., 2014): MALDER	Global dataset	Missingness > 0.1; KING kinship > 0.0884	2744	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025	425904
Pairwise FST	tidypopgen: pairwise_pop_fst - method = "WC84" (Weir and Cockerham, 1984)	Global dataset	Missingness > 0.1; KING kinship > 0.0884	2744	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	198196

Table 2: Supplementary Table 2: Minimum z-scores for f3 statistics across all combinations for each Malagasy population.

Malagasy	Asian	African	est	se	z	p
Betsileo	Borneo	DRC_Shi	-0.0115276	5.51e-05	-209.2158	0

Malagasy	Asian	African	est	se	z	p
Betsimisaraka	Borneo	DRC_Shi	-0.0110031	5.16e-05	-213.0884	0
Diego	Borneo	DRC_Shi	-0.0102260	4.85e-05	-210.7807	0
Merina	Borneo	DRC_Shi	-0.0104804	5.67e-05	-184.9265	0
Mikea	Borneo	DRC_Ngwi	-0.0099074	6.14e-05	-161.3010	0
Sakalava Tsimihety	Borneo	DRC_Shi	-0.0111734	5.08e-05	-220.1194	0
Temoro	Borneo	DRC_Shi	-0.0110101	5.37e-05	-205.2034	0
Vezo	Borneo	DRC_Shi	-0.0109646	5.46e-05	-200.6620	0

Table 3: Supplementary Table 3: f4 ratios calculated with admixtools. f4 ratios were calculated in the configurations f4(GBR,Chimpanzee;Malagasy,Borneo) / f4(GBR,Chimpanzee;Esan,Borneo) and f4(GBR,Chimpanzee;Malagasy,Esan) / f4(GBR,Chimpanzee;Borneo,Esan)

pop1	pop2	pop3	pop4	pop5	alpha	se	z
GBR	chimp_ref	Betsimisaraka	Borneo	ESN	0.5946328	0.0090260	65.87983
GBR	chimp_ref	MerinaA	Borneo	ESN	0.2431837	0.0120343	20.20760
GBR	chimp_ref	MerinaB	Borneo	ESN	0.5501539	0.0133817	41.11243
GBR	chimp_ref	Diego	Borneo	ESN	0.6109459	0.0090605	67.42992
GBR	chimp_ref	Sakalava_Tsimihety	Borneo	ESN	0.5839059	0.0089299	65.38787
GBR	chimp_ref	Betsileo	Borneo	ESN	0.4564963	0.0095466	47.81746
GBR	chimp_ref	Temoro	Borneo	ESN	0.5708479	0.0099019	57.65021
GBR	chimp_ref	Vezo	Borneo	ESN	0.5554886	0.0101537	54.70806
GBR	chimp_ref	Mikea	Borneo	ESN	0.5831392	0.0115808	50.35376

Table 4: Supplementary Table 4: MALDER results for all Malagasy populations, with varied African and Asian source populations. Columns are as follows: ...

test_res	p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	admixture_date	CI	amp_exp	amp_exp_CI	amp_aff
success	4.6e-222	Betsileo	Borneo	Esan	31.81	26.05 +/- 0.80	0.00216033 +/- 0.00006791	26.05	0.80	0.0021603	6.79e-05	0.0000299
success	3.6e-234	Betsileo	Borneo	Yoruba	32.67	25.97 +/- 0.78	0.00214609 +/- 0.00006568	25.97	0.78	0.0021461	6.57e-05	0.0000296
success	5.7e-230	Betsileo	Borneo	Mozambique_Chopi	32.38	26.44 +/- 0.82	0.00227784 +/- 0.00006829	26.44	0.82	0.0022778	6.83e-05	0.0000310
success	3.5e-188	Betsileo	Sumba	Esan	29.26	26.26 +/- 0.82	0.00191612 +/- 0.00006549	26.26	0.82	0.0019161	6.55e-05	0.0000261
success	1.7e-196	Betsileo	Sumba	Yoruba	29.91	26.18 +/- 0.80	0.00190193 +/- 0.00006360	26.18	0.80	0.0019019	6.36e-05	0.0000258
success	1.5e-200	Betsileo	Sumba	Mozambique_Chopi	30.22	26.69 +/- 0.84	0.00202856 +/- 0.00006714	26.69	0.84	0.0020286	6.71e-05	0.0000271
success	8.6e-200	Betsileo	Dai	Esan	30.16	26.21 +/- 0.87	0.00216652 +/- 0.00006781	26.21	0.87	0.0021665	6.78e-05	0.0000304
success	1.5e-205	Betsileo	Dai	Yoruba	30.59	26.14 +/- 0.85	0.00215206 +/- 0.00006551	26.14	0.85	0.0021521	6.55e-05	0.0000300
success	1.3e-197	Betsileo	Dai	Mozambique_Chopi	29.99	26.60 +/- 0.89	0.00228532 +/- 0.00006899	26.60	0.89	0.0022853	6.90e-05	0.0000315
success	1.1e-196	Betsimisaraka	Borneo	Esan	29.92	28.87 +/- 0.96	0.00187297 +/- 0.00004607	28.87	0.96	0.0018730	4.61e-05	0.0000083
success	7.2e-204	Betsimisaraka	Borneo	Yoruba	30.47	28.94 +/- 0.95	0.00186972 +/- 0.00004541	28.94	0.95	0.0018697	4.54e-05	0.0000081
success	6.9e-217	Betsimisaraka	Borneo	Mozambique_Chopi	31.43	28.84 +/- 0.92	0.00197274 +/- 0.00004997	28.84	0.92	0.0019727	5.00e-05	0.0000086
success	4.5e-185	Betsimisaraka	Sumba	Esan	29.01	28.90 +/- 1.00	0.00165692 +/- 0.00005122	28.90	1.00	0.0016569	5.12e-05	0.0000072
success	2.2e-189	Betsimisaraka	Sumba	Yoruba	29.35	28.96 +/- 0.99	0.00165296 +/- 0.00005075	28.96	0.99	0.0016530	5.08e-05	0.0000070
success	1.6e-193	Betsimisaraka	Sumba	Mozambique_Chopi	29.68	28.83 +/- 0.97	0.00175133 +/- 0.00005520	28.83	0.97	0.0017513	5.52e-05	0.0000074
success	1.3e-199	Betsimisaraka	Dai	Esan	30.14	28.81 +/- 0.96	0.00186998 +/- 0.00004787	28.81	0.96	0.0018700	4.79e-05	0.0000080
success	6.6e-205	Betsimisaraka	Dai	Yoruba	30.54	28.88 +/- 0.95	0.00186641 +/- 0.00004697	28.88	0.95	0.0018664	4.70e-05	0.0000078
success	3.4e-204	Betsimisaraka	Dai	Mozambique_Chopi	30.49	28.76 +/- 0.94	0.00196995 +/- 0.00005151	28.76	0.94	0.0019699	5.15e-05	0.0000083
success	1.5e-144	Diego	Borneo	Esan	25.60	20.26 +/- 0.65	0.00146813 +/- 0.00005735	20.26	0.65	0.0014681	5.74e-05	0.0000493
success	4.1e-148	Diego	Borneo	Yoruba	25.92	20.17 +/- 0.64	0.00145881 +/- 0.00005628	20.17	0.64	0.0014588	5.63e-05	0.0000494
success	4.4e-159	Diego	Borneo	Mozambique_Chopi	26.87	20.26 +/- 0.64	0.00153534 +/- 0.00005713	20.26	0.64	0.0015353	5.71e-05	0.0000514
success	2.2e-116	Diego	Sumba	Esan	22.93	20.57 +/- 0.73	0.00131657 +/- 0.00005741	20.57	0.73	0.0013166	5.74e-05	0.0000432
success	9.8e-120	Diego	Sumba	Yoruba	23.27	20.46 +/- 0.71	0.00130741 +/- 0.00005619	20.46	0.71	0.0013074	5.62e-05	0.0000433
success	6.7e-124	Diego	Sumba	Mozambique_Chopi	23.67	20.53 +/- 0.72	0.00137919 +/- 0.00005826	20.53	0.72	0.0013792	5.83e-05	0.0000452
success	4.7e-130	Diego	Dai	Esan	24.26	20.24 +/- 0.66	0.00146933 +/- 0.00006055	20.24	0.66	0.0014693	6.06e-05	0.0000496
success	1.1e-133	Diego	Dai	Yoruba	24.60	20.14 +/- 0.64	0.00145999 +/- 0.00005934	20.14	0.64	0.0014600	5.93e-05	0.0000497
success	2.3e-138	Diego	Dai	Mozambique_Chopi	25.04	20.23 +/- 0.64	0.00153750 +/- 0.00006140	20.23	0.64	0.0015375	6.14e-05	0.0000517
success	1.2e-228	Merina	Borneo	Esan	32.28	22.65 +/- 0.70	0.00176093 +/- 0.00005003	22.65	0.70	0.0017609	5.00e-05	0.0004215
success	5.5e-236	Merina	Borneo	Yoruba	32.80	22.66 +/- 0.69	0.00175499 +/- 0.00004855	22.66	0.69	0.0017550	4.86e-05	0.0004206
success	1.5e-226	Merina	Borneo	Mozambique_Chopi	32.13	23.18 +/- 0.72	0.00186187 +/- 0.00004419	23.18	0.72	0.0018619	4.42e-05	0.0004401
success	1.1e-227	Merina	Sumba	Esan	32.21	22.68 +/- 0.65	0.00154560 +/- 0.00004798	22.68	0.65	0.0015456	4.80e-05	0.0003678
success	2.2e-238	Merina	Sumba	Yoruba	32.97	22.68 +/- 0.64	0.00153947 +/- 0.00004669	22.68	0.64	0.0015395	4.67e-05	0.0003670
success	1.6e-250	Merina	Sumba	Mozambique_Chopi	33.81	23.32 +/- 0.69	0.00164251 +/- 0.00004452	23.32	0.69	0.0016425	4.45e-05	0.0003852
success	3.8e-233	Merina	Dai	Esan	32.60	22.74 +/- 0.70	0.00176475 +/- 0.00005140	22.74	0.70	0.0017648	5.14e-05	0.0004208
success	3.9e-244	Merina	Dai	Yoruba	33.37	22.75 +/- 0.68	0.00175932 +/- 0.00004984	22.75	0.68	0.0017593	4.98e-05	0.0004199
success	1.8e-241	Merina	Dai	Mozambique_Chopi	33.19	23.29 +/- 0.70	0.00186752 +/- 0.00004541	23.29	0.70	0.0018675	4.54e-05	0.0004395
success	5.5e-141	MerinaA	Borneo	Esan	25.28	21.70 +/- 0.86	0.00172501 +/- 0.00006674	21.70	0.86	0.0017250	6.67e-05	0.0000510
success	2.5e-139	MerinaA	Borneo	Yoruba	25.13	21.69 +/- 0.86	0.00172051 +/- 0.00006620	21.69	0.86	0.0017205	6.62e-05	0.0000510
success	1.8e-135	MerinaA	Borneo	Mozambique_Chopi	24.77	21.86 +/- 0.88	0.00180736 +/- 0.00006513	21.86	0.88	0.0018074	6.51e-05	0.0000546
success	1.8e-128	MerinaA	Sumba	Esan	24.11	21.96 +/- 0.86	0.00151669 +/- 0.00006290	21.96	0.86	0.0015167	6.29e-05	0.0000432
success	4.2e-129	MerinaA	Sumba	Yoruba	24.17	21.94 +/- 0.86	0.00151185 +/- 0.00006254	21.94	0.86	0.0015118	6.25e-05	0.0000433
success	8.1e-137	MerinaA	Sumba	Mozambique_Chopi	24.90	22.17 +/- 0.89	0.00159416 +/- 0.00006327	22.17	0.89	0.0015942	6.33e-05	0.0000466
success	1.4e-149	MerinaA	Dai	Esan	26.05	21.80 +/- 0.84	0.00173212 +/- 0.00006550	21.80	0.84	0.0017321	6.55e-05	0.0000519
success	1.3e-149	MerinaA	Dai	Yoruba	26.05	21.80 +/- 0.84	0.00172795 +/- 0.00006479	21.80	0.84	0.0017280	6.48e-05	0.0000519
success	1.2e-146	MerinaA	Dai	Mozambique_Chopi	25.79	21.98 +/- 0.85	0.00181659 +/- 0.00006355	21.98	0.85	0.0018166	6.36e-05	0.0000556
success	2.3e-53	MerinaB	Borneo	Esan	15.38	26.42 +/- 1.72	0.00182359 +/- 0.00006434	26.42	1.72	0.0018236	6.43e-05	0.0000235
success	1.2e-58	MerinaB	Borneo	Yoruba	16.15	26.46 +/- 1.64	0.00181557 +/- 0.00006173	26.46	1.64	0.0018156	6.17e-05	0.0000232
success	4.2e-61	MerinaB	Borneo	Mozambique_Chopi	16.49	26.93 +/- 1.63	0.00193439 +/- 0.00005846	26.93	1.63	0.0019344	5.85e-05	0.0000234
success	5.1e-52	MerinaB	Sumba	Esan	15.18	26.11 +/- 1.72	0.00159222 +/- 0.00006515	26.11	1.72	0.0015922	6.52e-05	0.0000201
success	8.6e-57	MerinaB	Sumba	Yoruba	15.88	26.13 +/- 1.65	0.00158453 +/- 0.00006284	26.13	1.65	0.0015845	6.28e-05	0.0000198
success	8.8e-60	MerinaB	Sumba	Mozambique_Chopi	16.31	26.75 +/- 1.64	0.00170122 +/- 0.00005956	26.75	1.64	0.0017012	5.96e-05	0.0000200
success	1.9e-49	MerinaB	Dai	Esan	14.78	26.44 +/- 1.79	0.00181330 +/- 0.00006328	26.44	1.79	0.0018133	6.33e-05	0.0000236
success	4.9e-54	MerinaB	Dai	Yoruba	15.48	26.48 +/- 1.71	0.00180611 +/- 0.00006077	26.48	1.71	0.0018061	6.08e-05	0.0000233
success	3.8e-57	MerinaB	Dai	Mozambique_Chopi	15.93	26.94 +/- 1.69	0.00192369 +/- 0.00005553	26.94	1.69	0.0019237	5.55e-05	0.0000236
success	6.2e-63	Mikea	Borneo	Esan	16.74	24.43 +/- 1.46	0.00179737 +/- 0.00007812	24.43	1.46	0.0017974	7.81e-05	0.0000009
success	4.2e-61	Mikea	Borneo	Yoruba	16.49	24.45 +/- 1.48	0.00179523 +/- 0.00007745	24.45	1.48	0.0017952	7.74e-05	0.0000008
success	8.6e-54	Mikea	Borneo	Mozambique_Chopi	15.44	23.99 +/- 1.55	0.00184531 +/- 0.00007972	23.99	1.55	0.0018453	7.97e-05	0.0000014

Table 4: Supplementary Table 4: MALDER results for all Malagasy populations, with varied African and Asian source populations. Columns are as follows: ... *(continued)*

test_res	p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	admixture_date	CI	amp_exp	amp_exp_CI	amp_aff
success	6.7e-59	Mikea	Sumba	Esan	16.18	24.75 +/- 1.53	0.00158879 +/- 0.00007863	24.75	1.53	0.0015888	7.86e-05	0.0000010
success	1.3e-56	Mikea	Sumba	Yoruba	15.86	24.77 +/- 1.56	0.00158630 +/- 0.00007822	24.77	1.56	0.0015863	7.82e-05	0.0000009
success	6.3e-50	Mikea	Sumba	Mozambique_Chopi	14.86	24.30 +/- 1.64	0.00163606 +/- 0.00008013	24.30	1.64	0.0016361	8.01e-05	0.0000015
success	2.2e-58	Mikea	Dai	Esan	16.11	24.68 +/- 1.53	0.00181233 +/- 0.00008078	24.68	1.53	0.0018123	8.08e-05	0.0000010
success	1.1e-56	Mikea	Dai	Yoruba	15.87	24.70 +/- 1.56	0.00181012 +/- 0.00008012	24.70	1.56	0.0018101	8.01e-05	0.0000009
success	2.4e-50	Mikea	Dai	Mozambique_Chopi	14.92	24.24 +/- 1.62	0.00186109 +/- 0.00008254	24.24	1.62	0.0018611	8.25e-05	0.0000015
success	1.1e-70	Sakalava_Tsimihety	Borneo	Esan	17.77	22.42 +/- 1.26	0.00177727 +/- 0.00007455	22.42	1.26	0.0017773	7.46e-05	0.0000396
success	4.3e-69	Sakalava_Tsimihety	Borneo	Yoruba	17.57	22.33 +/- 1.27	0.00176631 +/- 0.00007391	22.33	1.27	0.0017663	7.39e-05	0.0000395
success	3.3e-70	Sakalava_Tsimihety	Borneo	Mozambique_Chopi	17.71	23.09 +/- 1.30	0.00188537 +/- 0.00007484	23.09	1.30	0.0018854	7.48e-05	0.0000422
success	1.4e-73	Sakalava_Tsimihety	Sumba	Esan	18.14	22.70 +/- 1.25	0.00157199 +/- 0.00007657	22.70	1.25	0.0015720	7.66e-05	0.0000344
success	3.5e-71	Sakalava_Tsimihety	Sumba	Yoruba	17.84	22.60 +/- 1.27	0.00156087 +/- 0.00007598	22.60	1.27	0.0015609	7.60e-05	0.0000343
success	1.2e-73	Sakalava_Tsimihety	Sumba	Mozambique_Chopi	18.15	23.48 +/- 1.29	0.00167853 +/- 0.00007755	23.48	1.29	0.0016785	7.76e-05	0.0000367
success	1.4e-72	Sakalava_Tsimihety	Dai	Esan	18.02	22.60 +/- 1.25	0.00177662 +/- 0.00007776	22.60	1.25	0.0017766	7.78e-05	0.0000398
success	3.9e-71	Sakalava_Tsimihety	Dai	Yoruba	17.83	22.51 +/- 1.26	0.00176524 +/- 0.00007692	22.51	1.26	0.0017652	7.69e-05	0.0000396
success	5.3e-72	Sakalava_Tsimihety	Dai	Mozambique_Chopi	17.94	23.26 +/- 1.30	0.00188434 +/- 0.00007879	23.26	1.30	0.0018843	7.88e-05	0.0000423
success	9.6e-133	Temoro	Borneo	Esan	24.52	29.11 +/- 0.86	0.00194337 +/- 0.00007926	29.11	0.86	0.0019434	7.93e-05	0.0000182
success	1e-128	Temoro	Borneo	Yoruba	24.14	29.07 +/- 0.90	0.00193347 +/- 0.00008011	29.07	0.90	0.0019335	8.01e-05	0.0000178
success	7.4e-147	Temoro	Borneo	Mozambique_Chopi	25.81	29.35 +/- 0.87	0.00205756 +/- 0.00007973	29.35	0.87	0.0020576	7.97e-05	0.0000186
success	5.3e-104	Temoro	Sumba	Esan	21.66	29.34 +/- 0.87	0.00172607 +/- 0.00007970	29.34	0.87	0.0017261	7.97e-05	0.0000165
success	2.7e-101	Temoro	Sumba	Yoruba	21.37	29.28 +/- 0.90	0.00171615 +/- 0.00008032	29.28	0.90	0.0017162	8.03e-05	0.0000162
success	7.5e-112	Temoro	Sumba	Mozambique_Chopi	22.47	29.60 +/- 0.86	0.00183343 +/- 0.00008158	29.60	0.86	0.0018334	8.16e-05	0.0000169
success	5.4e-130	Temoro	Dai	Esan	24.26	29.48 +/- 0.82	0.00195981 +/- 0.00008079	29.48	0.82	0.0019598	8.08e-05	0.0000181
success	6.1e-127	Temoro	Dai	Yoruba	23.97	29.42 +/- 0.85	0.00194883 +/- 0.00008131	29.42	0.85	0.0019488	8.13e-05	0.0000178
success	2.8e-142	Temoro	Dai	Mozambique_Chopi	25.40	29.71 +/- 0.81	0.00207391 +/- 0.00008166	29.71	0.81	0.0020739	8.17e-05	0.0000185
success	1.9e-69	Vezo	Borneo	Esan	17.62	21.98 +/- 1.25	0.00184944 +/- 0.00006413	21.98	1.25	0.0018494	6.41e-05	0.0000285
success	5.5e-72	Vezo	Borneo	Yoruba	17.94	22.08 +/- 1.23	0.00184557 +/- 0.00006245	22.08	1.23	0.0018456	6.24e-05	0.0000284
success	1.2e-72	Vezo	Borneo	Mozambique_Chopi	18.03	22.26 +/- 1.24	0.00194431 +/- 0.00005711	22.26	1.24	0.0019443	5.71e-05	0.0000301
success	2.4e-67	Vezo	Sumba	Esan	17.34	22.16 +/- 1.28	0.00163475 +/- 0.00005995	22.16	1.28	0.0016347	6.00e-05	0.0000246
success	1.6e-70	Vezo	Sumba	Yoruba	17.76	22.26 +/- 1.25	0.00163029 +/- 0.00005825	22.26	1.25	0.0016303	5.82e-05	0.0000245
success	1.3e-70	Vezo	Sumba	Mozambique_Chopi	17.76	22.41 +/- 1.26	0.00172420 +/- 0.00005415	22.41	1.26	0.0017242	5.42e-05	0.0000261
success	4.2e-66	Vezo	Dai	Esan	17.17	22.04 +/- 1.28	0.00185290 +/- 0.00006451	22.04	1.28	0.0018529	6.45e-05	0.0000280
success	1.1e-68	Vezo	Dai	Yoruba	17.51	22.13 +/- 1.26	0.00184873 +/- 0.00006279	22.13	1.26	0.0018487	6.28e-05	0.0000278
success	3.5e-70	Vezo	Dai	Mozambique_Chopi	17.71	22.29 +/- 1.26	0.00194813 +/- 0.00005747	22.29	1.26	0.0019481	5.75e-05	0.0000296

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